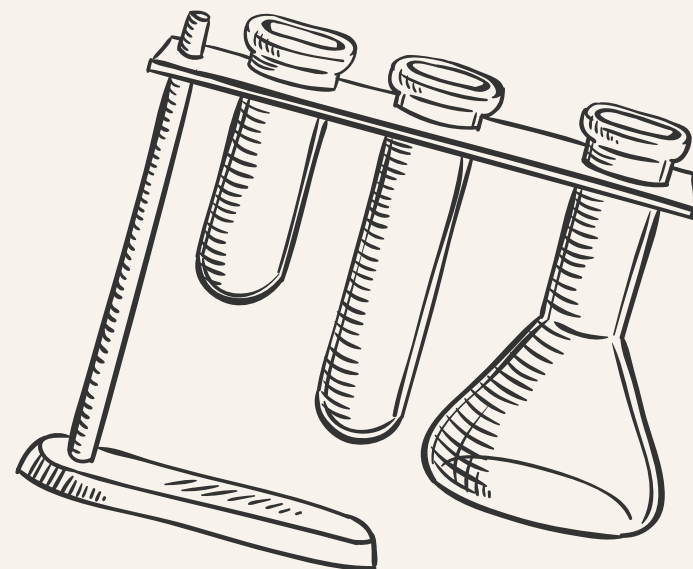
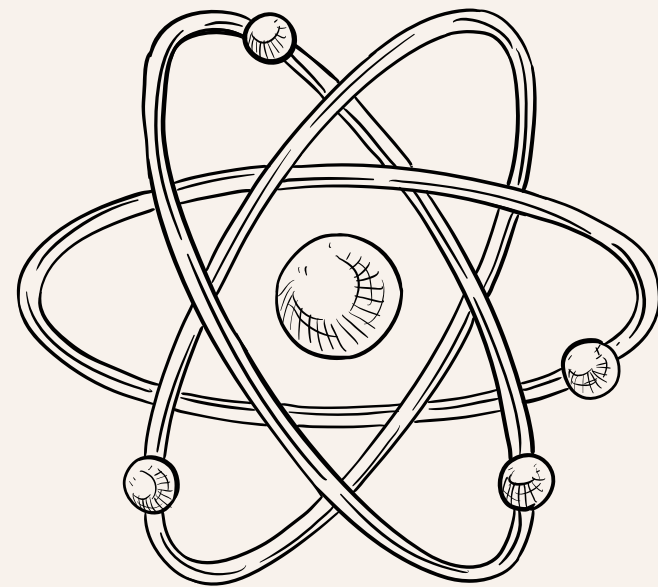
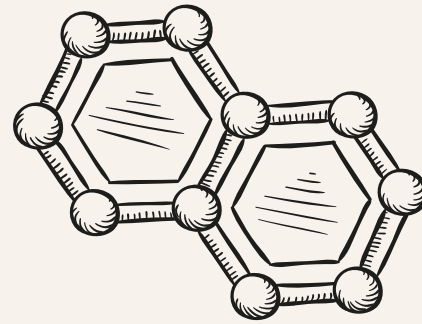
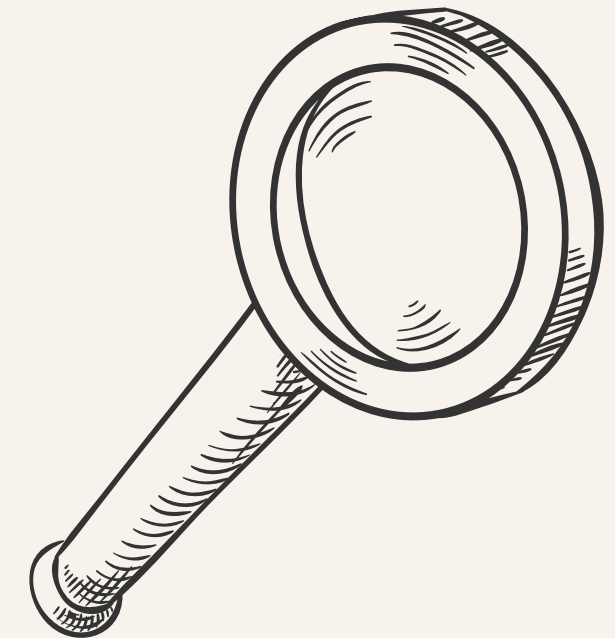
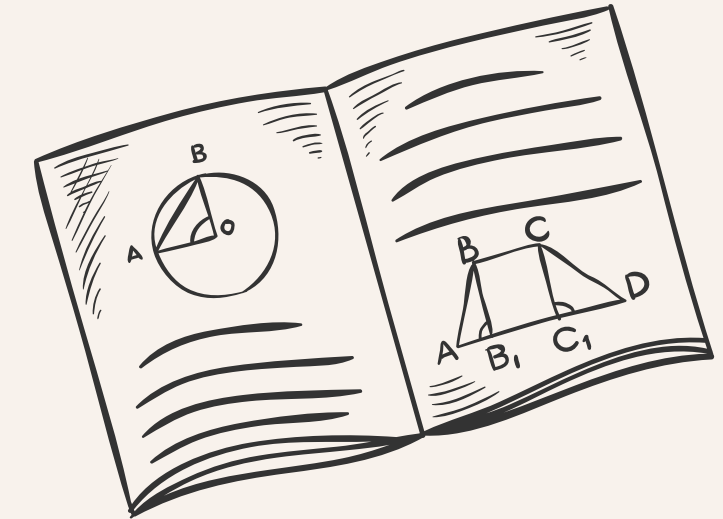


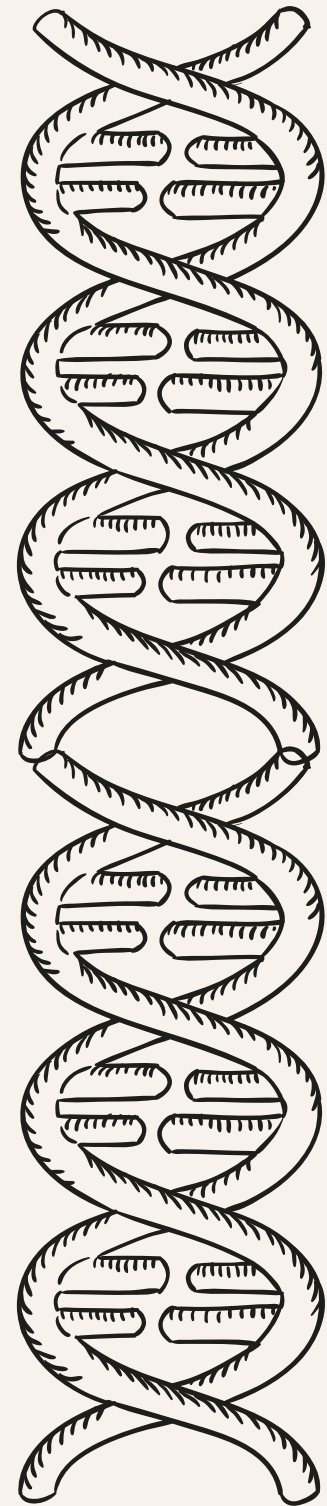


Evolutionary Computation

JENETICS: EVOLUTIONARY COMPUTATION IN JAVA

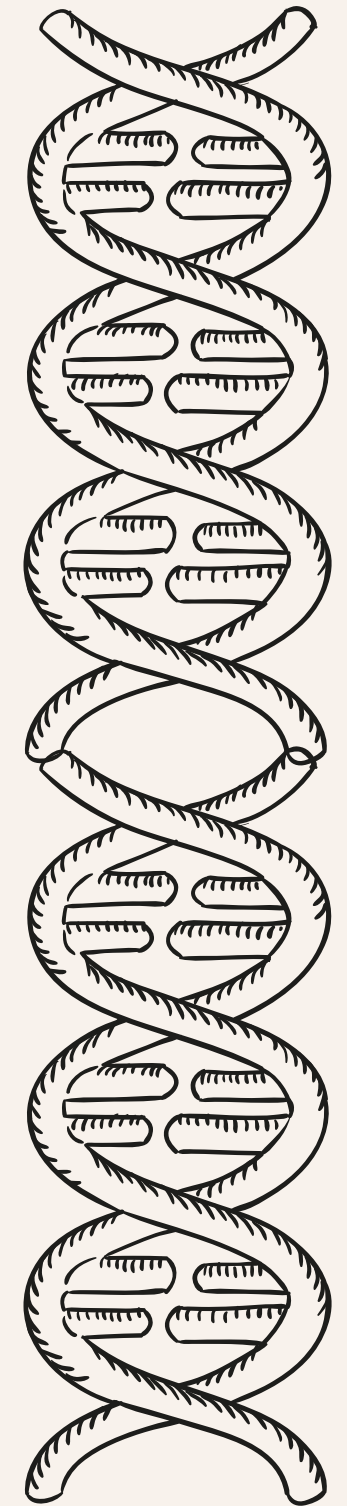
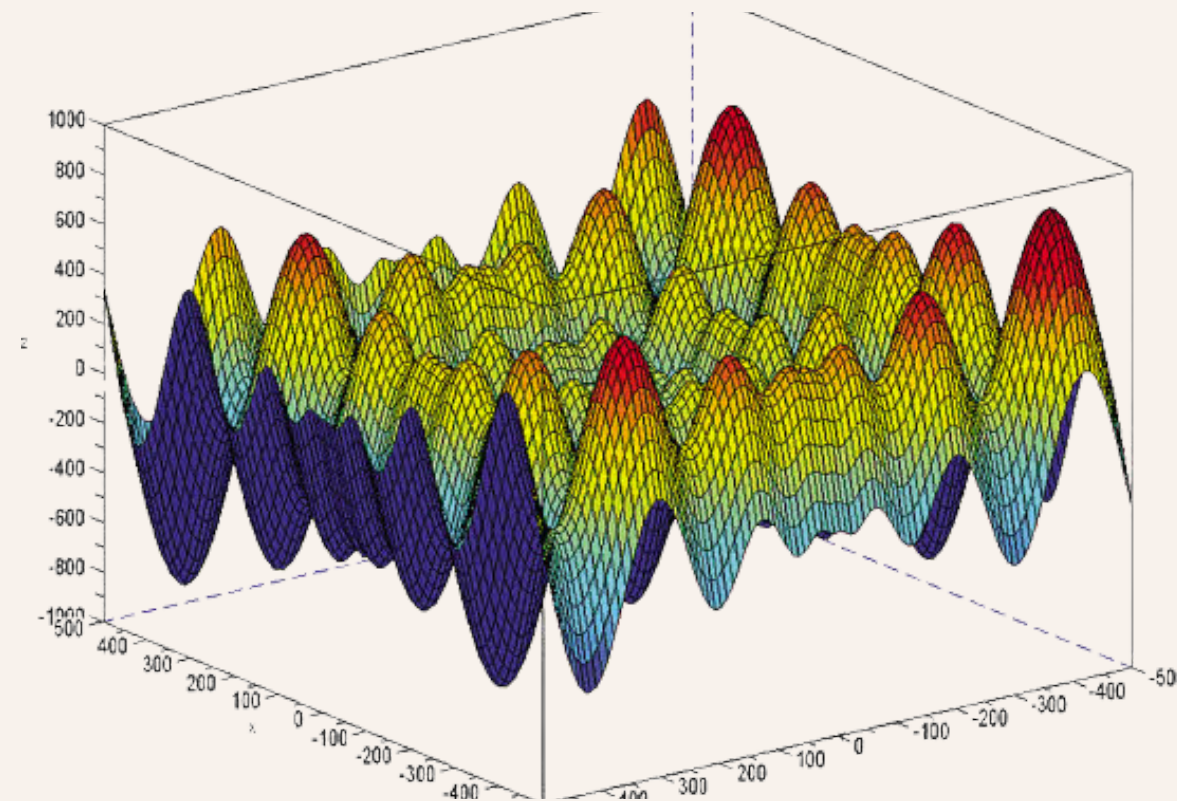
Presented by: Caleb Odunade

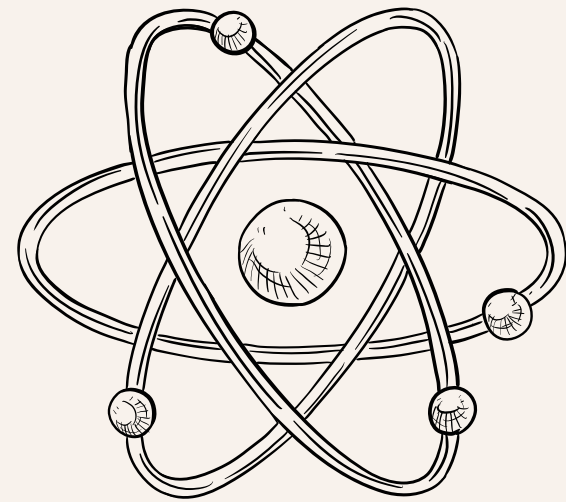




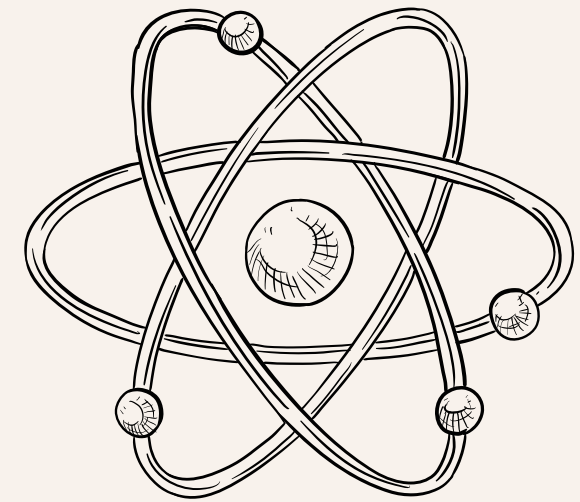
What is Evolutionary Optimization?

To automatically find near-optimal solutions to complex problems by mimicking the biological process of natural selection





What is Jenetics?



- Java-Based
 - Built specifically for Java applications
 - Traditional OOP
- Structured Tool Set
 - Chromosomes and genes
 - Selection methods
 - Crossover operators
 - Mutation operators
 - Fitness evaluation
- Optimization
 - Built specifically for Java applications
 - Integrates easily with existing Java projects
- Mimics Natural Selection
 - Creates a population of possible solutions
 - Selects stronger solutions
 - Applies crossover and mutation
 - Improves solutions over generations

Jenetics v. DEAP

- Stronger type safety
 - Jenetics uses Java generics, so many structural errors are caught at compile time instead of runtime.
- Clearer GA structure
 - Jenetics classes like Gene, Chromosome, Genotype, Phenotype, and Engine make the algorithm easier to read and follow.
- Better fit for Java projects
 - Jenetics integrates directly into existing Java applications without adding a separate Python environment.
- DEAP is more flexible
 - DEAP is lightweight and easy to customize, while Jenetics is more structured and organized.



Knapsack Problem

Input: Items with a weight and value

Constraint: Total weight must stay under capacity

Goal: Maximize total value

EA: Represent Each item as a Bit

1: Take Item

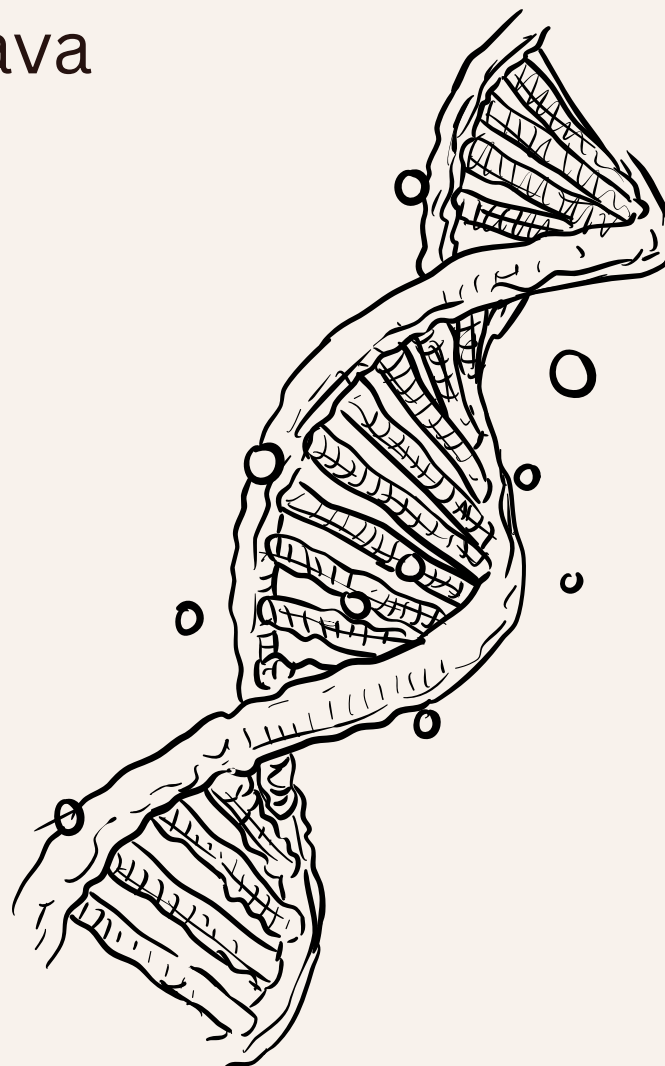
0: Leave Item



You must pick a combination of items to maximize total value without going over a weight limit

Initialization

01 Download Extension Pack for Java



02 Maven pom.xml

```
<dependency>
  <groupId>io.jenetics</groupId>
  <artifactId>jenetics</artifactId>
  <version>9.1.0</version>
</dependency>

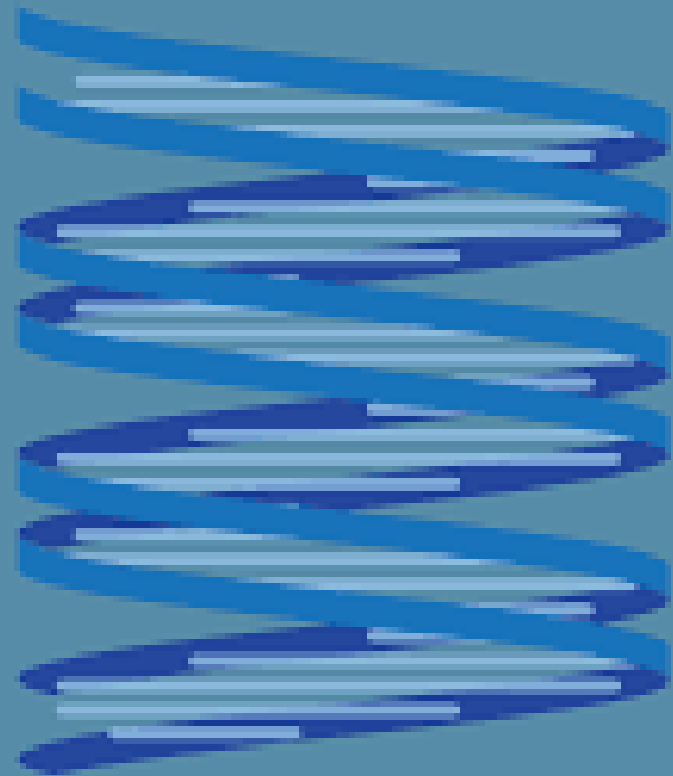
....

<plugin>
  <groupId>org.codehaus.mojo</groupId>
  <artifactId>exec-maven-plugin</artifactId>
  <version>3.6.4</version>
  <configuration>

<mainClass>com.example.demo.DemoKnapsack</mai
nClass>
  </configuration>
</plugin>
```

JENETICS

LIBRARY USER'S MANUAL



BLANC WISBLAM EDITOR

VERSION 1.1

Demo Time!